Kshetrajna

Private adaptive intelligence for Windows

A local assistant that learns app routines and explains its suggestions.

Team SankatMochan

Hackathon prototype v0.3

The problem

Daily work spans apps, system resources and files.

Resource pressure

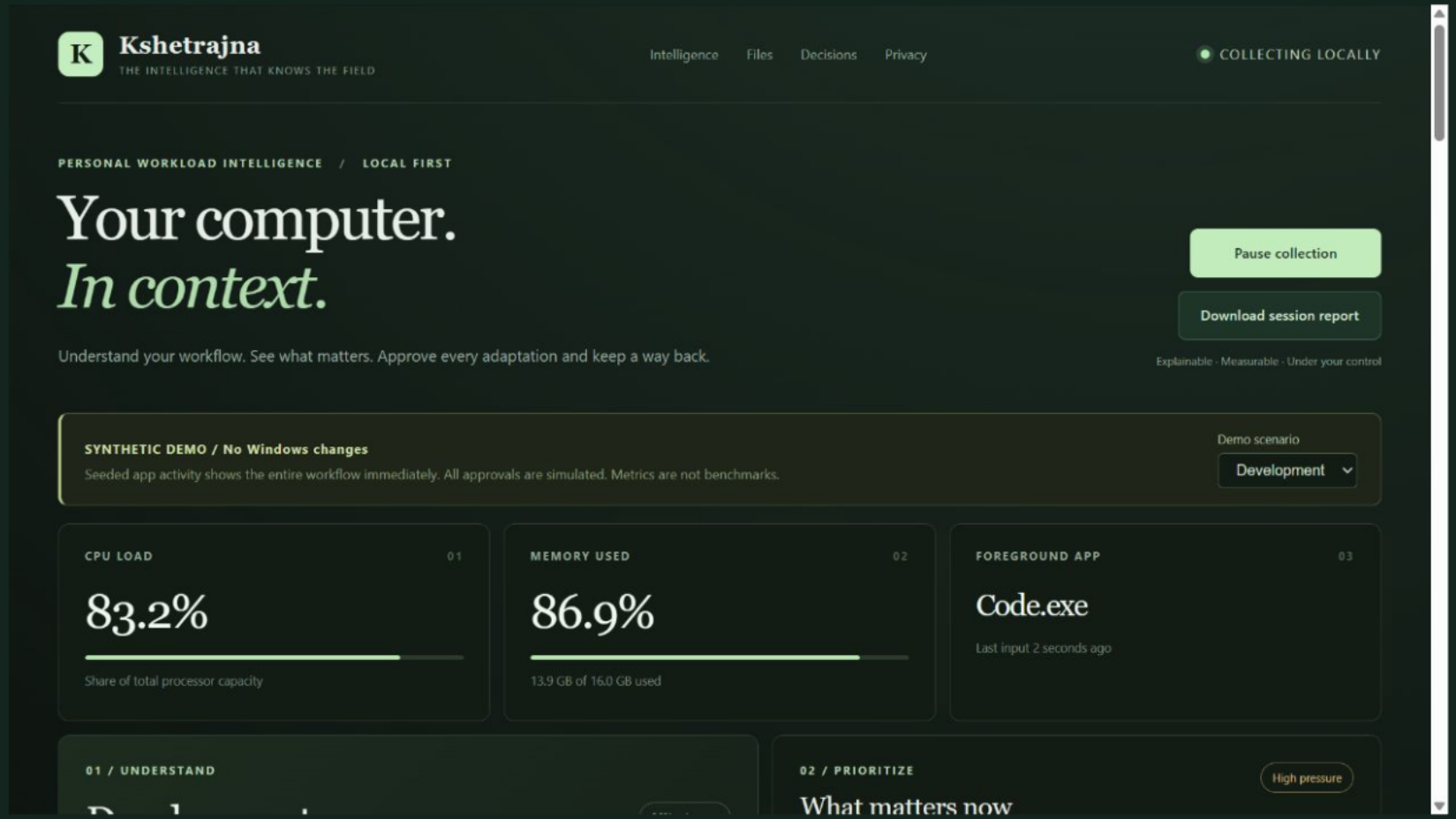
Users inspect CPU and RAM separately from what they are trying to do.

Scattered context

The files relevant to a workflow sit apart from app activity and system tools.

Kshetrajna brings those signals into one local dashboard.

The working dashboard



System view

CPU, RAM and foreground activity

Personal context

Workflow routines and relevant local files

Screenshot: synthetic demo. Live observation requires user consent.

Workflow and file intelligence

App routines

Recognizes recurring app groups and the order in which apps appear.

Local file catalog

Indexes approved folders with metadata and bounded text excerpts.

Next-app suggestions

Uses recent session history and withholds weak predictions.

Relevant files

Offers keyword search and explains why a file matches the current context.

Reviewed move/rename API includes a journal and conflict-aware undo.

Consent and privacy

User-controlled collection

Live observation starts only after opt-in. Pause, retention and deletion controls remain visible.

Local processing

Python and SQLite run on the laptop. The dashboard binds to loopback.

Reviewed resource trials

Eligible CPU-priority changes require approval and expire after 60 seconds.

Recorded outcomes

The journal keeps approvals and undo status. Before/after deltas are observations.

Telemetry excludes keystrokes, URLs and window titles. File excerpts require approved folders.

Windows ARM64 architecture

Python 3.13	Workflow rules and session-based prediction
Win32 via ctypes	CPU/RAM, processes and foreground activity
SQLite	Local telemetry, file catalog and action journals
HTML / CSS / JS	Browser dashboard at 127.0.0.1

Test device: HP Snapdragon X (X126100), 16 GB RAM, ARM64

All 61 tests passed on the Snapdragon ARM64 laptop. The NPU is not used.

Validation and next steps

61

tests passed on the Snapdragon ARM64 laptop

0

third-party runtime packages

Next: device diagnostics, organizer UI and controlled performance trials.

Demo: run_demo.cmd Live mode: run.cmd

https://github.com/Team-SankatMochan/KSHETRAJNA-OS